QUANT

Permutation, Combination & Probability

Assignment 05

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P & C Assignments 05

1. In a chess tournament each contestant plays once against each of the other contestants. If in all 45 games are played, then the number of participants is -

1. 9 2.10 3. 15. 4. None of these

2. The number of ways in which 12 books can be put in 3 shelves, 4 on each shelf is

1. $\frac{12!}{(4!)^3}$ 2. $\frac{12!}{3!\,(4!)^3}$ 3. $\frac{12!}{(3!)^3\,4!}$ 4. None of these

3. The total number of ways in which 12 persons can be divided into three groups of 4–persons each is

1. $\frac{12!}{(3!)^3\,4!}$ 2. $\frac{12!}{(4!)^3}$ 3. $\frac{12!}{(4!)^3\,3!}$ 4. $\frac{12!}{(3!)^4}$

4. There are n–white and n–black balls marked 1, 2, 3_____n. The number of ways in which we can arrange these balls in a row so that neighbouring balls are of different colours is -

1. $n!$ 2. (2n)! 3. $2.(n!)^2$ 4. $\frac{(2n)!}{(n!)^2}$

5. A and B play a game where each is asked to select a number form 1 to 25. If the two numbers match, both of them win a prize. The probability that they will not win a prize in a single trial is -

1. $\frac{1}{25}$ 2. $\frac{24}{25}$ 3. $\frac{2}{25}$ 4. None of these

6. The probability that a man will live 10 more years is $\frac{1}{4}$ and the probability that his wife will live 10 more years is $\frac{1}{3}$. The probability that neither of them will be alive in 10 years is -

1. $\frac{5}{12}$ 2. $\frac{1}{2}$ 3. $\frac{7}{12}$ 4. $\frac{11}{12}$

7. The probability that a person will hit a target in shooting practice is 0.3. If the person shoots 10 times, the probability that he hits the target is -

1. 1 2. $1-(0.7)^{10}$ 3. $(0.7)^{10}$ 4. $(0.3)^{10}$

8. A box contains 10 mangoes out of which 4 are rotten. 2 mangoes are taken out together. If one of them is found to be good, the probability that the other is also good is -

1. $\frac{1}{3}$ 2. $\frac{8}{15}$ 3. $\frac{5}{13}$ 4. $\frac{2}{13}$

9. A box contains 24 identical balls of which 12 are white and 12 black. The balls are drawn at random from the box, one at a time with replacement. The probability that a white ball is drawn for the 4th time on the 7th draw is -

1. $\frac{5}{64}$ 2. $\frac{27}{32}$ 3. $\frac{5}{32}$ 4. $\frac{1}{2}$

10. The probability that a student will succeed in entering B-School '*X*' is 0.2 and the probability that he will succeed in entering B-School '*Y*' is 0.5. If the probability that he will be successful in entering both B-Schools is 0.3, then the probability that he does not succeed at both places is

1. 0.4 2. 0.3 3. 0.2 4. 0.6

Answer key & Explanations

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1.		If there n – participants $\Rightarrow {}^{n}C_2 = 45 \Rightarrow \frac{n(n-1)}{2} = 45 \Rightarrow n = 10$
2.		Here the shelves are different 4 – Books on 1st Shelf in for 12 Books $= {}^{12}C_4 = \frac{12!}{4!8!}.4 -$ Books on 2nd shelf the remaining 8 Books $= {}^{8}C_4 = \frac{8!}{4!4!}$. 4 Books on 3rd shelf for the 4 – books $= {}^{4}C_4 = \frac{4!}{4!O!} = 1$. Required. No. $\frac{12!}{4!8!} \times \frac{8!}{4!4!} = \frac{12!}{(4!)^3}$
3.		Here mere grouping is of concern. Thus Required No. of ways $= \frac{12!}{(4!)^3 3!}$
4.		We can arrange n white and n black balls alternatively in the following ways (i) WB WB———, (ii) BW BW———, So required. number $= n! \times n! + n!n! = 2(n!)^2$
5.		The total number of ways of choosing number from 1 to 25 by both the players $= 25 \times 25 = 625$. There are 25 ways in which number chosen by both the players is same. The probability that they will prize in a single trial $= \frac{25}{625} = \frac{1}{25}$. Required. Probability that they will not win a prize in a single trial $= 1 - \frac{1}{25} = \frac{24}{25}$
6.		Reqd. Probability $= \left(1 - \frac{1}{4}\right)\left(1 - \frac{1}{3}\right) = \frac{1}{2}$.
7.		Reqd. Prob. $= 1 -$ Prob. that he does not hit the target in any trial. $= 1 - (0.7)^{10}$
8.		A = Event that first mango is good. B = Event that second mango is good. Reqd. Prob. $= P(B/A) = \frac{P(A \cap B)}{P(A)}$. $P(A \cap B) =$ Prob. that both the mangoes are good $= \frac{{}^{6}C_2}{{}^{10}C_2}.P(A) =$ Prob. that first mango is good $= \frac{{}^{6}C_2}{{}^{10}C_2} + \frac{{}^{6}C_1 \times {}^{4}C_1}{{}^{10}C_2}$. Hence $P\left(\frac{B}{A}\right) = \frac{{}^{6}C_2}{{}^{6}C_2 + {}^{6}C_1 \times {}^{4}C_1} = \frac{5}{13}$.
9.		$P_1 =$ prob. of drawing a white ball. $P_2 =$ prob. Of drawing a black ball. $P_1 = P_2 = \frac{1}{2}$. White ball is drawn 4th time in 7th draw, means that three white balls have been drawn in first six draws and 4th white balls is drawn in 7th draw. Reqd. prob $= \left({}^{6}C_3 P_1^3 P_2^3\right) P_1 = {}^{6}C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)$
10.		A = Event that he is selected in B–schools 'x', B = Event that he is selected in B–schools 'y' $P(A) = 0.2, P(B) = 0.5, P(A \cap B) = 0.3$

		Reqd. Probability $= P\left(\bar{A} \cap \bar{B}\right) = 1 - P\left(A \cup B\right)$ $= 1 - [P\left(A\right) + P\left(B\right) - P\left(A \cap B\right)]$, = 0.6